

Practical 2: From Values to Variables

In Practical 1 you examined whether COMPAS produced fair outcomes across racial groups. Today you'll apply the same critical lens to a different system: content moderation tools and NLP toxicity or harmful language classifiers. The difference is that this time you'll map the human values at stake *before* you touch the data, and then test whether those values are reflected (or violated) in what the data actually shows.

By the end of this session you will begun work on the Dataset Audit Table required for your assignment.

Content warning: The Jigsaw Toxicity dataset contains comments referencing identity groups in contexts that some students may find distressing. The CSV provided on blackboard title `sample_no_comment_text.csv` **does not include comment text**, only identity tags and toxicity labels. If you have concerns at any point, please speak to the session leader.

Part 1 Stakeholder Mapping [c. 45 minutes]

Individual Warm-Up [10 minutes]

Select one of the following short articles about content moderation to read, or find your own relating to an angle or identity group captured in the coursework dataset that interests you.

- <https://www.techpolicy.press/the-censorship-of-lgbtq-content-online-corresponds-with-declines-in-freedom-for-everyone/>
- <https://www.techpolicy.press/why-content-moderation-must-account-for-disability/>
- <https://techcrunch.com/2019/08/14/racial-bias-observed-in-hate-speech-detection-algorithm-from-google/>

As you read, keep these two questions in mind:

- Where does the content in this system come from, and how might that platform's culture shape what gets labelled as toxic?
- Who is harmed when the system makes a mistake?

Note down one observation from your article before moving on.

Small Group Task

Work in groups of between 3-5 students. Randomly assign each member one of the following stakeholder roles:

Role

- Marginalised community members using reclaimed language
- Content creator (general)

Role

- Platform trust & safety team
- Advertisers / platform business interests
- External auditor

Step 1: Identify a value and priority for your role [5 minutes]

Identify a value you think would be important to that stakeholder group and justify why you think it would be important. Draft a summary statement to encapsulate this, e.g. "My stakeholder most needs the system to _____, because _____."

Step 2: Are there any tensions? [10 minutes, as a group]

Each person reads their sentence aloud. The group can provide feedback and if there are factors you hadn't considered you can refine your statement.

Then as a group, identify:

- Which two stakeholders have the most direct conflict?
- Is there any stakeholder whose needs are simply invisible to the others, i.e. not even in conflict, just not considered?

Step 3: Name one concrete harm [10 minutes, as a group]

Agree on one specific harm, e.g. what may result from caused by a False Positive, i.e. a safe post incorrectly labelled as toxic or a False Negative, an unsafe post incorrectly labelled as not-toxic? Which stakeholder does this harm most? Which stakeholder barely notices?

How are the different stakeholder values impacted by this? What value tensions emerge when exploring these examples?

Write your group's sharpest value tension in one sentence:

E.g. "[Stakeholder X] values [...], but [Stakeholder Y] values [...], and the system cannot fully serve both."

Step4: Whole-Class Share [10 minutes]

Each group shares their value tension sentence.

Before moving to Part 2, look at all the tensions collected and as a group vote on which tension you think will show up most clearly in the data. Write your prediction here:

Our prediction: _____

Part 2: Empirical Audit

For the remainder of the session you can begin using the dataset provided for the assignment to consider how the stakeholder perspectives you have just explored may be present in the data.

Jupyter Notebook Tasks & Data — 45 minutes

Open the starter notebook on Blackboard in the Practicals folder, which guides you through the following steps. `practical_2_data_analysis_starter_notebook.ipynb`

Use the data file provided there for your analysis.

Key point: The `sample_no_comment_text.csv` file contains identity tags and toxicity labels (target column), but no comment text. You are auditing the *pattern of labels*, not reading offensive content.

Work through the following steps:

1. Load and familiarise yourself with the data structure.

2. Examine the class balance

Calculate the overall class distribution. What proportion of comments are labelled toxic vs. non-toxic?

Is this dataset balanced or imbalanced? What might be the implications of this for training a classifier?

3. Examine the subgroup distribution

Select **two identity categories** to compare. Choose your own based on your Part 1 discussion

Count how many comments are tagged with each identity category.

Are these groups represented equally in the dataset?

4. Compare Toxicity Rates

For each identity category, calculate the proportion of comments that are labelled toxic.

Create a bar chart comparing these two rates.

5. Connect Back to Part 1 – Discuss in pairs / small groups

Look at your group's prediction from Part 1.

- Does your finding support or challenge it?
- How confident are you in your findings? Why?
- Share your findings with a partner. Together, agree on:
 - One finding you're confident in
 - One tentative recommendation - what should be changed, and at which stage (data collection, annotation guidance, model training)?

6. Next steps

Reflect on the representation gaps and empirical analysis you have conducted. Use this to begin drafting the audit table required in your assignment.